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U.S Congress members' trading activities: A case of NANC and KRUZ

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ABSTRACT

This paper analyzes U.S. Congress members' investment behaviors through ETFs NANC and KRUZ, reflecting Democratic and Republican trading patterns. While NANC outperforms KRUZ, neither significantly outperforms market returns, suggesting regulatory measures like the STOCK Act mitigate informational advantages.

1. Introduction

In 2011, a significant episode of 60 Min and a publication by Peter Schweizer (Schweizer, 2011) provided anecdotal evidence that members of Congress and their families engaged in trading based on non-public information. This ignited increased public scrutiny of Congress members' investing and trading activities. Over the past 3 years, there were 1946 filings by 213 politicians for 3370 issuers with a total volume of $2.037 billion. For example, Ro Khanna (Democrat) traded over 14,700 times in 1165 different stocks listed across 43 countries. Ninety-six Democrats and 116 Republicans have traded actively over the past 3 years. The most popular stocks include: Apple (43 members of Congress), Alphabet (40 members), Microsoft (40 members), and Amazon (39 members). In terms of the most traded stock, Microsoft was traded 445 times, Amazon, 411 times, and Alphabet, 317 times while the U.S. T-bills were traded 297 times over the past 3 years. The resulting effect is the creation of two ETFs: KRUZ and NANC, which track the trading activities of Congress members.

Theoretically, investing in the same stocks as members of Congress should lead to abnormal returns if members of Congress are trading on private information. However, the introduction of the STOCK Act aimed at addressing insider trading by members of Congress and their families by requiring members of Congress to disclose their financial transactions can reduce potential abnormal returns. Furthermore, politicians do align their investment choices with their ideological positions to some extent. Legislators with conservative ideologies tend to invest more heavily in industries that align with conservative policy positions while liberal-leaning politicians show preferences for industries associated with progressive policies such as socially responsible investing (Aiken et al., 2020).

Our results demonstrate a strong relationship between the investment choices of U.S. Congress members and their political affiliations, reflected through the performance of NANC and KRUZ ETFs. NANC, closely associated with Democratic members, shows a strong correlation with technology-heavy indices like NASDAQ, achieving an annual return of 27 %. In contrast, KRUZ, mirroring Republican members' preferences, correlates more with the Dow Jones Industrial Average, with a comparatively lower annual return of 13 %. Despite these differences, neither ETF significantly outperforms the market on a risk-adjusted basis, suggesting that the STOCK Act effectively neutralizes the advantages of insider knowledge. These findings imply that while political affiliations influence investment patterns, regulatory measures might have mitigated the exploitation of non-public information.

The remainder of the paper is as follows: in Section 2 we review the literature, in Section 3 we outline the data, in Section 4 we present the results and in Section 5, we conclude the paper.

2. Literature review

The empirical evidence suggests that individuals can earn abnormal returns (4.9 % over three months) by simply utilizing a buy-and-hold strategy following senators' trades (Hanousek et al., 2023). Similar findings are reported by Karadas (2018) and Berkman et al., (2014). Furthermore, Tahoun (2014) shows that congressional stock ownership

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affects legislative decision-making and the allocation of political favors. Abdurakhmonov et al., (2023) show that trading in stocks disclosed by senators generates statistically significant abnormal returns when the senator has jurisdiction over the firm. Examining whether politicians receive stock tips from brokerage houses, Stephan et al., 2021 show that trades by politicians who are politically connected to the brokerage house where the trade is executed are more profitable. Connected trades earn an incremental 0.3 % over a five-day window compared to the politician's average profitability.

In contrast, Eggers and Hainmueller, 2013 analyze the trades of Congress members in the 2004–2008 period. Their findings suggest that congressional stock trading strategies do not yield superior returns and may even result in losses compared to passive investment strategies. They show that the poor performance persists across different political parties and time periods, indicating a consistent pattern of subpar investment decisions among legislators. However, they show that members of Congress invest disproportionately in local firms and campaign contributors. These connected investments (and particularly local investments) generally outperform members' other investments (Eggers and Hainmueller, 2014).

The STOCK Act aimed at addressing insider trading by members of Congress and their families by requiring members of Congress to disclose their financial transactions. Empirical evidence appears to support the impact of the STOCK Act in reducing trading on private information by politicians and their families. In fact, Karadas (2018) shows that politicians' spouses traded on time-sensitive, value-relevant information between 2004 and 2010, generating short-term abnormal returns exceeding 12 % at a 1-week holding period and 6 % at a 3-month holding period on an annual basis on their portfolios influenced by the politicians' committee assignments. However, portfolios of politicians and their spouses underperformed the market during the 2011–2014 period, suggesting a change in trading behavior due to public pressure and the implementation of the STOCK Act (Karadas, 2018). Similarly, Huang and Xuan, (2023) find that the Act led to a decrease in trading activity among senators, suggesting that it effectively deterred them from engaging in potentially questionable financial activities. They observe a decrease in the profitability of senatorial trades after the Act's passage, indicating a reduction in the exploitation of private information for personal gain. Overall, the paper provides empirical evidence supporting the efficacy of the STOCK Act in promoting transparency and ethical conduct among lawmakers.

Given the mixed evidence that Congress members earn abnormal returns from trading securities using privileged information and the passage of the STOCK Act aimed at promoting transparency and ethical conduct among lawmakers, the introduction of NANC and KRUZ with the aim of tracking trading activities of U.S Congress members and their families warrants an investigation.

3. Data and methodology

3.1. Data

We obtain daily and weekly stock returns for NANC and KRUZ for the period February 1, 2023-January 31, 2024. from Yahoo Finance. We extract returns for the S&P 500, NASDAQ, Dow Jones, and several ETFs tracking Aerospace and Defence stocks and U.S Oil prices from Yahoo Finance. We use a 52-week risk-free rate of 4.89 % to calculate our Sharpe ratio. Further, we obtain Fama-French-Carhart five-factors: market factor (MKTRF), size factor (SMB), value factor (HML), momentum factor (MOM), and investment factor (CMA), from WRDS database for the same time period.

3.2. Methodology

We utilized a modified Fama-French-Carhart five-factor model to examine whether NANC and KRUZ generate risk adjusted abnormal returns. We estimate EQ (1) below using daily returns.

$$Ret_t = \alpha + \beta MKT_t + \gamma SMB_t + \delta HML_t + \mu MOM_t + \theta CMA_t + \varepsilon_t, \tag{1}$$

Where *Ret* is the daily returns for either NANC or KRUZ ETF, MKT is the risk adjusted market returns, SMB is the size factor, HML is the growth factor (Fama and French, 1993), UMD is the momentum factor (Carhart, 1997) and CMA is the conservative factor.[1]

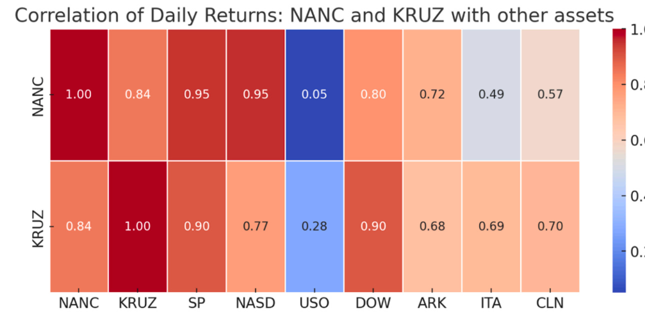

Fig. 1. Daily returns correlation.
Note: USO is an exchange-traded security that invests primarily in futures contracts for light, sweet crude oil, other types of crude oil, diesel-heating oil, gasoline, natural gas, and other petroleum-based fuels. ARKK (ARK Innovation) is an ETF that will invest under normal circumstances primarily (at least 65 % of its assets) in domestic and foreign equity securities of companies that are relevant to the fund's investment theme of disruptive innovation. **ITA is the** iShares U.S. Aerospace & Defense ETF. CLN is **the** iShares Global Clean Energy ETF.

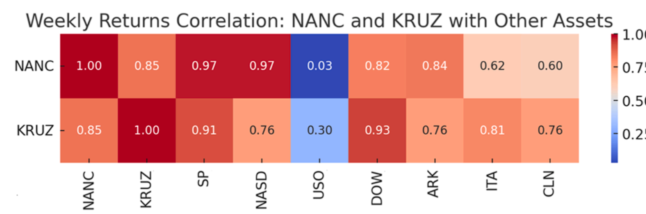

Fig. 2. Weekly returns correlation.
Note: USO is an exchange-traded security that invests primarily in futures contracts for light, sweet crude oil, other types of crude oil, diesel-heating oil, gasoline, natural gas, and other petroleum-based fuels. ARKK (ARK Innovation) is an ETF that will invest under normal circumstances primarily (at least 65 % of its assets) in domestic and foreign equity securities of companies that are relevant to the fund's investment theme of disruptive innovation. ITA is the iShares U.S. Aerospace & Defense ETF. CLN is the iShares Global Clean Energy ETF.

4. Results

In Fig. 1, we report the Pearson correlation coefficients for NANC and KRUZ with several indices. The correlation between NANC and KRUZ is 0.84. NANC seems to be more correlated with S&P 500 (0.95) and the NASDAQ (0.95) compared to KRUZ which is more correlated with the S&P 500 (0.90). KRUZ tends to be more correlated with US Oil ETF (0.28) compared to NANC (0.05). Finally, KRUZ is more correlated with the DOW (0.90) compared to NANC. The correlation patterns are primarily driven by the core holdings. Also, KRUZ is correlated with the Aerospace and Defence ETF (0.69) and Global Clean Energy ETF (0.70). Surprisingly, NANC's correlation with the Global Clean Energy ETF is 0.57. Finally, the weekly return data results in a similar correlation matrix (Fig. 2).

[1] Note: we did not include liquidity factor as it is updated annually and the most up-to date data ends December 2023.

Table 1
KRUZ Top holdings.

	KRUZ-TOP 10 HOLDINGS					
	% OF NET ASSETS	NAME	TICKER	CUSIP	SHARES HELD	MARKET VALUE
1	4.20 %	JP Morgan Chase & Co.	JPM	46625H100	7873	1947,465.28
2	3.74 %	Comfort Systems USA Inc	FIX	199,908,104	3492	1735,524.00
3	3.48 %	NVIDIA Corp	NVDA	67066G104	11,334	1614,414.96
4	2.46 %	Simon Property Group Inc	SPG	828,806,109	6282	1141,690.68
5	2.39 %	AT&T Inc	T	00206R102	46,424	1108,605.12
6	2.30 %	Accenture PLC	ACN	G1151C101	2936	1064,887.20
7	2.28 %	Fidelity National Information Services Inc	FIS	3 1620M106	12,410	1054,601.80
8	2.25 %	Allstate Corp/The	ALL	020,002,101	5130	1044,673.20
9	2.20 %	Chevron Corp	CVX	166,764,100	6570	1019,926.80
10	2.13 %	Arista Networks Inc	ANET	040,413,205	9112	986,374.00
…..	….	…..	….	….	…..	…..
161	0.09 %	AstraZeneca PLC	AZN	46,353,108	634	43,238.80

Panel B: KRUZ HOLDINGS SECTOR ALLOCATION

Industry	Percentage holdings
Technology	20.87 %
Industrials	20.59 %
Financials	15.81 %
Consumer Services	8.89 %
Health Care	8.17 %
Oil & Gas	7.37 %
Basic Materials	5.82 %
Consumer Goods	4.60 %
Utilities	2.82 %
Telecommunications	2.70 %

Note: Top 10 holdings by KRUZ as of 12/09/2024. Holdings change based on the trading activities of Republican Congress members and their family. https://www.subversiveetfs.com/kruz.
Note: Sectorial breakdown as of 09/30/2024. Holdings change based on the trading activities of Republican Congress members and their family. https://www.marketwatch.com/investing/fund/kruz/holdings.

Table 2
NANC Top 10 holdings and industry breakdown.

Panel A: Top 10 holdings

	NANC-TOP 10 HOLDINGS					
	% OF NET ASSETS	NAME	TICKER	CUSIP	SHARES HELD	MARKET VALUE $
1	12.51 %	NVIDIA Corp	NVDA	67066G104	182,101	25,938,466.44
2	7.90 %	Microsoft Corp	MSFT	594,918,104	36,944	16,387,250.08
3	5.11 %	Amazon.com Inc	AMZN	023,135,106	46,652	10,591,403.56
4	5.05 %	Salesforce Inc	CRM	79466L302	28,901	10,461,872.99
5	4.10 %	Apple Inc	AAPL	037,833,100	35,041	8509,356.44
6	3.51 %	Alphabet Inc	GOOG	02079K107	41,245	7271,081.05
7	3.06 %	American Express Co	AXP	025,816,109	20,876	6346,095.24
8	3.00 %	Costco Wholesale Corp	COST	22 160K105	6273	6226,642.53
9	2.82 %	Netflix Inc	NFLX	641 10L106	6245	5837,451.30
10	2.64 %	Philip Morris International Inc	PM	718,172,109	41,949	5476,861.44
…..	…..	……	….	…..	…..	……
167	0.0002 %	Magnera Corp	MAGN	55939A107	21	414.96

Panel B: NANC HOLDINGS SECTOR ALLOCATION

Industry Sector	Percentage Holdings
Technology	46.84 %
Consumer Services	16.28 %
Health Care	10.31 %
Financials	8.74 %
Consumer Goods	8.03 %
Industrials	8.01 %
Non-Classified Equity	0.60 %
Telecommunications	0.44 %
Basic Materials	0.42 %
Oil & Gas	0.10 %

Note: Top 10 holdings by NANC as of 12/09/2024. Holdings change based on the trading activities of Democratic Congress members and their family. https://www.subversiveetfs.com/nanc.
Note: Sectoral breakdown as of 09/30/2024. Holdings change based on the trading activities of Democratic Rongress members and their family. https://www.marketwatch.com/investing/fund/kruz/holdings.

Table 3
Performance.

	Volatility		Returns		Sharpe Ratio
	Weekly	Annualized	Weekly	Annualized	Weekly
NANC	0.021	0.151	0.0051	0.2655	1.43
KRUZ	0.020	0.143	0.0025	0.1285	0.56
S&P 500	0.018	0.132	0.0037	0.1924	1.09
NASDAQ	0.024	0.171	0.0055	0.2858	1.39
USO	0.042	0.305	0.0019	0.1014	0.17
DOW	0.017	0.121	0.0025	0.1276	0.65
ARKK	0.058	0.417	0.0042	0.2202	0.41
ITA	0.020	0.144	0.0019	0.0976	0.34
CLN	0.036	0.263	−0.0055	−0.2882	−1.28

Note: USO is an exchange-traded security that invests primarily in futures contracts for light, sweet crude oil, other types of crude oil, diesel-heating oil, gasoline, natural gas, and other petroleum-based fuels. ARKK (ARK Innovation) is an ETF that will invest under normal circumstances primarily (at least 65 % of its assets) in domestic and foreign equity securities of companies that are relevant to the fund's investment theme of disruptive innovation. ITA is the iShares U.S. Aerospace & Defense ETF. CLN is the iShares Global Clean Energy ETF.

In Table 1, Panel A, only 2 of the top ten holdings in KRUZ are technology firms with Nvidia Corp accounting for 3.74 % of the total assets under management for KRUZ. The technology sector makes up 21 % of total holdings, followed by industrials (21 %), financials (16 %), consumer services (9 %), and health care (8 %) (Table 1, panel A). In comparison, 6 out of the top 10 holdings in NANC are technology firms with Nvidia Corp accounting for 12.51 % of the assets under management, followed by Microsoft (7.9 %) and Amazon (5.11 %) (Table 2, Panel A). In fact, all of the top 6 holdings in NANC are technology firms. The technology sector makes up 47 % of the entire portfolio, followed by consumer services at 16.28 %, healthcare at 10.31 % and financials at 8.74 % (Table 2, panel B). Even though both ETFs' total number of holdings are similar (NANC =167 and KRUZ=161 companies), NANC is more concentrated in the technology sector.

In Table 3, we report the performance and volatility. The annualized return for NANC is 27 % with an annualized volatility of 15 %. KRUZ on the other hand, has an annualized return of 13 % and a volatility of 14 %. This is not unexpected given the top 10 holdings for both ETFs. Since NANC is more technology-oriented, its returns are more closely aligned with the NASDAQ while slightly underperforming compared to the NASDAQ (28.6 %). However, its Sharpe ratio is higher (1.43) than the NASDAQ (1.39).

In terms of KRUZ, its performance is like the Dow Jones index (12.76 %). However, its Sharpe ratio (0.56) is lower than the Dow (0.65). Compared to the S&P 500 and NASDAQ, KRUZ significantly underperforms both in terms of annualized returns and Sharpe ratio.

Our results are consistent with the argument that the STOCK Act has reduced trading on private information by politicians and their families and hence, they do not outperform the market. Our findings are consistent with Karadas (2018) and Huang and Xuan, (2023) who find a decrease in the profitability of senatorial trades after the STOCK Act's passage, indicating a reduction in the exploitation of privileged information for personal gain.

4.1. Regression results

The regression results from the modified Fama-French-Carhart five factor model are reported in Table 4. In column I-IV we report the results of NANC. The Alpha is not significantly different from zero in any of the models. This suggests that an ETF tracking member of congress and their family trading does not produce superior risk adjusted returns. This supports our Sharpe ratio analysis above and is consistent with the argument that the STOCK Act reduced trading on private information by politicians and their families and hence, investing in an ETF tracking trading activities of politicians and their families does not lead to abnormal returns.

In columns X-IX we report the results for KRUZ. The results are similar to those of NANC. The alpha is not statistically different from zero in any of the models.

5. Conclusion and discussion

This study aims to explore the investment behaviors of U.S. Congress members and their market performance through the lens of two exchange-traded funds (ETFs), NANC and KRUZ, which track the trading activities of Democratic and Republican Congress members, respectively. The creation of these ETFs was premised on the hypothesis that legislators might possess or act upon material non-public information, potentially allowing investors to achieve abnormal returns by mimicking their trades.

Our empirical analysis spanned daily and weekly stock returns from February 1, 2023, to January 31, 2024, comparing the performance and volatility of NANC and KRUZ with major indexes and other relevant ETFs. We observed that NANC, heavily invested in technology stocks, aligned with the investment trends of Democratic members, reflecting a preference for sectors associated with progressive policies. Conversely, KRUZ, with significant holdings in industrials, aerospace, and defense stocks, mirrored the investment preferences of Republican members which are traditionally aligned with conservative policies.

The results underscored a high correlation between NANC and major technology-driven indexes like NASDAQ, whereas KRUZ showed a closer alignment with the Dow Jones Industrial Average, highlighting

Table 4
Modified Fama-French five factor model.

	NANC				KRUZ			
	I	II	III	IV	V	VI	VII	VIII
Alpha	0.0002 (0.0002)	0.0001 (0.0002)	0.0001 (0.0002)	0.0000 (0.0002)	−0.0002 (0.0002)	−0.0001 (0.0001)	−0.0001 (0.0001)	−0.0000 (0.0001)
MKT	1.0717*** (0.0221)	1.0905*** (0.0211)	1.0901*** (0.0212)	1.0201*** (0.0225)	0.9075*** (0.0242)	0.8795*** (0.0181)	0.8804*** (0.0182)	0.9146*** (0.0204)
SMB		−0.0648** (0.0257)	−0.0679** (0.0282)	−0.0515* (0.0263)		0.0952*** (0.0221)	0.1015*** (0.0242)	0.0934*** (0.0238)
HML		−0.1353*** (0.0249)	−0.1358*** (0.0250)	−0.0398 (0.0276)		0.2482*** (0.0214)	0.2491*** (0.0215)	0.2023*** (0.0250)
UMD			−0.0061 (0.0226)	0.0028 (0.0210)			0.0122 (0.0194)	0.0079 (0.0190)
CMA				−0.3006*** (0.0469)				0.1468*** (0.0425)
Adj R^2	0.9029	0.9217	0.9214	0.9323	0.8479	0.9243	0.9241	0.9273
NOBS	253	253	253	253	253	253	253	253

Note: The dependent variables in columns 1–1 V are the returns for NANC and in columns V-VIII are the returns for KRUZ ETF. MKT is the risk adjusted market returns, SMB is the size factor, HML is the growth factor (Fama-French, 1993), UMD is the momentum factor (Carhart, 1997) and CMA is the conservative factor.

the sectorial biases rooted in the political ideologies of the respective ETFs. Notably, the annualized return for NANC significantly outperformed KRUZ, with a return of 27 % compared to 13 % which could be attributed to the robust performance of the technology sector over the period analyzed.

Furthermore, the analysis revealed that both ETFs did not significantly outperform the market once adjusted for risk, as measured by the Sharpe ratio, suggesting that while there may be sector-specific gains, the broader market efficiency remains intact, limiting the ability to achieve sustained abnormal returns. This finding aligns with the intended effects of the STOCK Act, which has seemingly curbed the advantages of trading based on privileged information among politicians and their associates.

In conclusion, while our study provides evidence supporting the sectorial investment strategies aligned with political ideologies, it also highlights the challenges of leveraging political insider trading for consistent market outperformance. The regulatory environment, underscored by the STOCK Act, appears effective in mitigating undue advantages previously speculated to exist among political insiders. Investors considering strategies based on political trading activities should remain cautious, as the compliance landscape and market dynamics continue to evolve.

Data availability

Data will be made available on request.

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